PRACTICE EXAM 4 OF 4 · EXAM STRATEGY - ALL DOMAINS

Practice exam 4

The final twenty questions, then a trend review across all four papers.

QUESTIONS 20 TIME About 30 minutes MODULE 35 of 35

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> An auditor asks which control plane component is allowed to read and write etcd directly.

Which component is it?

A The kube-apiserver

B The kube-scheduler

C The kubelet

D The kube-controller-manager

◆ PAUSE AND DECIDE

Which component is it?

A The kube-apiserver
The only component that talks to etcd.

B The kube-scheduler
Works through the API server.

C The kubelet
Works through the API server.

D The kube-controller-manager
Works through the API server.

THE SIGNAL

Only the API server talks to etcd.

> Apps must reach a legacy database at db.legacy.example.com through an in-cluster name, so moving it later only changes one object.

What should the team create?

A A headless Service with a selector

B A NodePort Service

C An ExternalName Service

D A ClusterIP Service with a selector

◆ PAUSE AND DECIDE

What should the team create?

A A headless Service with a selector

Selects in-cluster Pods; there are none.

B A NodePort Service

Brings outside traffic in; wrong direction.

C An ExternalName Service

A DNS alias for the outside hostname.

D A ClusterIP Service with a selector

Needs Pods in the cluster to select.

THE SIGNAL

In-cluster name for an outside host: ExternalName.

> Quayside's payment worker processes a message queue. It must scale on queue length, and down to zero when the queue is empty.

Which tool fits?

A An HPA on the workers' CPU

B KEDA

C A VerticalPodAutoscaler

D Knative Serving

◆ PAUSE AND DECIDE

Which tool fits?

A An HPA on the workers' CPU

Cannot reach zero alone; CPU lags the queue.

B KEDA

Event-driven scaling, including to zero.

C A VerticalPodAutoscaler

Resizes Pods; never removes them.

D Knative Serving

Built for HTTP services, with revisions.

THE SIGNAL

Queue length, down to zero: KEDA.

> An engineer lists which kinds of object do not belong to any namespace.

Which of these is cluster-scoped?

A A Pod

B A ConfigMap

C A Role

D A Node

◆ PAUSE AND DECIDE

Which of these is cluster-scoped?

A A Pod
Namespaced.

B A ConfigMap
Namespaced.

C A Role
Namespaced; ClusterRole is the cluster-scoped one.

D A Node
Nodes belong to the whole cluster.

THE SIGNAL

Nodes, PVs, Namespaces, ClusterRoles: cluster-scoped.

> Security requires every call between Quayside's internal services to be encrypted, with both sides authenticated, and no code changes.

What meets the requirement?

A NetworkPolicies between services

B TLS termination at the Ingress

C Headless Services for every app

D A service mesh enforcing mutual TLS

◆ PAUSE AND DECIDE

What meets the requirement?

A NetworkPolicies between services
Restricts traffic; encrypts nothing.

B TLS termination at the Ingress
Covers traffic coming in only.

C Headless Services for every app
Changes DNS; no encryption.

D A service mesh enforcing mutual TLS
Encrypts and authenticates each call.

THE SIGNAL

Encrypt and authenticate internal calls: mesh mTLS.

> A CronJob's manifest was deleted from Git. Argo CD syncs automatically, but the CronJob still exists in the cluster.

Which setting makes Argo CD delete it?

A selfHeal: true

B prune: true

C A new targetRevision

D A Replace=true sync option

◆ PAUSE AND DECIDE

Which setting makes Argo CD delete it?

A selfHeal: true

Reverts edits; does not delete removed objects.

B prune: true

Deletes objects removed from Git.

C A new targetRevision

Changes the tracked commit.

D A Replace=true sync option

Changes how objects are applied.

THE SIGNAL

Removed from Git, still in the cluster: prune.

> A Deployment has 4 replicas and default rolling update settings. A new image is rolled out.

What limits hold throughout the rollout?

A At most 5 Pods; at least 3 available

B At most 4 Pods; at least 4 available

C At most 6 Pods; at least 2 available

D At most 8 Pods; none need be available

◆ PAUSE AND DECIDE

What limits hold throughout the rollout?

A At most 5 Pods; at least 3 available
25% of 4 is 1 either way.

B At most 4 Pods; at least 4 available
Would need both settings at zero.

C At most 6 Pods; at least 2 available
That is 50%, not the default.

D At most 8 Pods; none need be available
Nothing like the defaults.

THE SIGNAL

Defaults: 25% surge, 25% unavailable.

> Quayside's cluster still uses the community Ingress NGINX controller, which is now retired. The team wants the recommended way forward for HTTP traffic into the cluster.

What should they move to?

A Stay on it; it still gets patches

B A NodePort Service for every app

C A Gateway API implementation

D ExternalName Services

◆ PAUSE AND DECIDE

What should they move to?

A Stay on it; it still gets patches
No patches since March 2026.

B A NodePort Service for every app
No routing, and awkward for users.

C A Gateway API implementation
The recommended successor.

D ExternalName Services
Aliases point outwards, not inwards.

THE SIGNAL

Ingress NGINX retired: Gateway API.

> An engineer runs kubectl run debug --image=busybox:1.37 on a current cluster.

What does the command create?

A A Deployment

B A single Pod

C A Job

D A ReplicaSet

◆ PAUSE AND DECIDE

What does the command create?

A A Deployment
Only in very old kubectl versions.

B A single Pod
kubectl run creates one bare Pod.

C A Job
Needs kubectl create job.

D A ReplicaSet
Deployments create those.

THE SIGNAL

kubectl run: one bare Pod.

> A CNCF incubating project has met every graduation requirement and applies to graduate.

Which body approves the move?

A The CNCF Governing Board

B The Linux Foundation's executive team

C The Kubernetes Steering Committee

D The CNCF Technical Oversight Committee

◆ PAUSE AND DECIDE

Which body approves the move?

A The CNCF Governing Board
Business and budget, not maturity.

B The Linux Foundation's executive team
Not responsible for CNCF maturity.

C The Kubernetes Steering Committee
Governs Kubernetes only.

D The CNCF Technical Oversight Committee
The TOC decides every maturity move.

THE SIGNAL

Project maturity: the TOC.

> Machine-learning Pods must run only on the GPU nodes, and every other Pod must stay off those nodes.

What configuration achieves both?

A A toleration on ML Pods, with no taint added

B A nodeSelector on ML Pods, with no taint anywhere

C Taint, plus toleration and affinity on ML Pods

D A high PriorityClass for the ML Pods

◆ PAUSE AND DECIDE

What configuration achieves both?

A A toleration on ML Pods, with no taint added

A toleration without a taint does nothing.

B A nodeSelector on ML Pods, with no taint anywhere

Others can still land on GPU nodes.

C Taint, plus toleration and affinity on ML Pods

Taint keeps others out; affinity pulls ML in.

D A high PriorityClass for the ML Pods

Changes order, not placement.

THE SIGNAL

Others out and ours in: taint plus affinity.

> A Pod stays Pending with: 0/4 nodes are available: 4 node(s) didn't match Pod's node affinity/selector.

What is the cause?

A No node has the labels the Pod requires

B The nodes lack enough free memory

C The Pod's image cannot be pulled

D The scheduler has stopped running

◆ PAUSE AND DECIDE

What is the cause?

A No node has the labels the Pod requires

Exactly what the event says.

B The nodes lack enough free memory

That would say Insufficient memory.

C The Pod's image cannot be pulled

Pulls happen after scheduling.

D The scheduler has stopped running

A dead scheduler writes no events.

THE SIGNAL

Read the FailedScheduling reason literally.

> A chart's values.yaml sets replicaCount: 2. The team installs it with -f prod.yaml, which sets 4. The install command also passes --set replicaCount=6.

How many replicas are created?

A 2

B 4

C 6

D None: Helm rejects the conflict

◆ PAUSE AND DECIDE

How many replicas are created?

A 2

Defaults have the lowest priority.

B 4

Files beat defaults, but not --set.

C 6

--set has the highest priority.

D None: Helm rejects the conflict

Overrides are normal.

THE SIGNAL

Defaults, then -f files, then --set.

> A helper container in the booking API Pod converts the app's custom metrics output into Prometheus format.

Which multi-container pattern is that?

A Adapter

B Ambassador

C Sidecar, in the pattern sense

D Init

◆ PAUSE AND DECIDE

Which multi-container pattern is that?

A Adapter

Reshapes the app's output for others.

B Ambassador

Proxies the app's outgoing calls.

C Sidecar, in the pattern sense

Extends or enhances the app generally.

D Init

Runs before the app, then exits.

THE SIGNAL

Reshaping output: adapter.

> On a brand-new kubeadm cluster, every node is NotReady and the CoreDNS Pods are Pending.

What is most likely missing?

A kube-proxy

B etcd

C metrics-server

D A CNI network plugin

◆ PAUSE AND DECIDE

What is most likely missing?

A kube-proxy
kubeadm installs it.

B etcd
The control plane is up.

C metrics-server
Affects top and autoscaling only.

D A CNI network plugin
kubeadm installs no network plugin.

THE SIGNAL

Fresh kubeadm, NotReady, CoreDNS Pending: no CNI.

> A new version of the payment worker must never run at the same time as the old one, even briefly.
>
> A short outage during each release is acceptable.

Which Deployment strategy fits?

A RollingUpdate with maxUnavailable: 0

B Recreate

C A canary at 5%

D RollingUpdate with default settings

◆ PAUSE AND DECIDE

Which Deployment strategy fits?

A RollingUpdate with maxUnavailable: 0
Guarantees overlap.

B **Recreate**
Stops all old Pods before starting new ones.

C A canary at 5%
Overlap by design.

D RollingUpdate with default settings
Still overlaps.

THE SIGNAL

Must never overlap: Recreate.

> Quayside's contract with a partner promises 99.5% monthly availability, with refunds if it is missed.

What is this promise?

A An SLA

B An SLO

C An SLI

D An error budget

◆ PAUSE AND DECIDE

What is this promise?

A **An SLA**
A contract with consequences.

B **An SLO**
An internal target.

C **An SLI**
A measurement.

D **An error budget**
What an SLO leaves over.

THE SIGNAL

A contract with consequences: SLA.

> Quayside's GitOps repository must include the database password Secret for the bookings service.
>
> Nothing readable may be committed.

What is the safest approach?

A Commit the Secret manifest; values are base64

B Seal or encrypt it, or reference an external store

C Put the password in a ConfigMap instead

D Store it in an annotation on the Deployment

◆ PAUSE AND DECIDE

What is the safest approach?

A Commit the Secret manifest; values are base64
Base64 is readable by anyone.

B Seal or encrypt it, or reference an external store
Sealed Secrets, SOPS or External Secrets.

C Put the password in a ConfigMap instead
Worse: plain text.

D Store it in an annotation on the Deployment
Worse: plain text, and visible.

THE SIGNAL

Secrets in GitOps: encrypted or external, never plain.

> A container spec requests cpu: 500m.

How much CPU is that?

A 500 CPU cores

B 500 megabytes

C 5% of one CPU

D Half of one CPU

◆ PAUSE AND DECIDE

How much CPU is that?

A 500 CPU cores

m means thousandths, not whole cores.

B 500 megabytes

CPU is not measured in bytes.

C 5% of one CPU

That would be 50m.

D Half of one CPU

500 millicores: half a core.

THE SIGNAL

1000m is one CPU.

> A NetworkPolicy must allow traffic only from worker Pods that are also in namespaces labelled team=payments.

How should its from section be written?

A Two entries: a namespaceSelector, then a podSelector

B A podSelector entry on its own

C One entry holding both selectors together

D A namespaceSelector entry on its own

◆ PAUSE AND DECIDE

How should its from section be written?

A Two entries: a namespaceSelector, then a podSelector

Two entries combine with OR: far too wide.

B A podSelector entry on its own

Only Pods in the policy's own namespace.

C One entry holding both selectors together

Selectors in one entry combine with AND.

D A namespaceSelector entry on its own

Every Pod in those namespaces.

THE SIGNAL

Both conditions at once: one entry, both selectors.

Score yourself by domain

DOMAIN	QUESTIONS	YOUR SCORE
Kubernetes Fundamentals (44%)	1, 4, 7, 9, 11, 14, 16, 19	of 8
Container Orchestration (28%)	2, 5, 8, 12, 15, 20	of 6
Application Delivery (16%)	6, 13, 18	of 3
Cloud Native Architecture (12%)	3, 10, 17	of 3

ON THE EXAM

Now add this to your first three papers: the trend is next.

Your trend across all four papers

DOMAIN	OUT OF	YOUR TOTAL	READY AT
Kubernetes Fundamentals	35		30 or more
Container Orchestration	22		19 or more
Application Delivery	13		11 or more
Cloud Native Architecture	10		8 or more
All four papers	80		68 or more (85%)

ON THE EXAM

A domain below its 'ready' line is your study plan. The total is only the headline.

Where to go back to

WEAK DOMAIN	REWATCH	THEN
Kubernetes Fundamentals	Modules 1 to 12, scenarios first	Module 30's first three tables
Container Orchestration	Modules 13 to 21, especially 14 and 21	Module 30's orchestration tables
Application Delivery	Modules 22 to 25, and the Domain 3 review	Module 30's delivery tables
Cloud Native Architecture	Modules 26 to 28	Module 29, what changed

ON THE EXAM

Rewatch the worked scenarios before the teaching: they show the decisions the exam tests.

Before you book

- **Every domain at or above its ready line,** and 68 or more of 80 overall.
- **Replay modules 29 and 30** in the final week.
- **Check cncf.io/projects** and the Linux Foundation exam page for anything new.
- **Run the secure-browser system check** days before, not minutes.

> ON THE EXAM
>
> You have now worked through 240 exam-style decisions. The real exam asks about 60.

That is the whole course. Good luck in the exam, and welcome to the cloud native community.